

Deep Typographical Layout and Vector Generation in MoonBit

Abstract

This paper presents the design and implementation of **moon-typst**, a zero-dependency typographical layout engine and multi-format vector compiler implemented in MoonBit. It supports two-dimensional mathematical layout including matrices, vectors, and limits, and high-fidelity output serialization to SVG, HTML5 and Adobe PDF 1.4.

1. Mathematical Kinetics and Tensor Formulation

Modern computational typesetting requires rigorous spatial bounding algorithms for complex mathematical

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

When projected into high-dimensional vector spaces, state vectors satisfy the continuity condition

$$\lim_{I \rightarrow \infty} \frac{x}{sum}$$

The asymptotic energy dispersion across frequency bands is evaluated via the limit series

With localized spectral estimation modulated by the accented field estimators $\hat{\theta}$ and $\tilde{\omega}$, baseline coherence is strictly maintained.

2. Implementation in Pure MoonBit

The compiler architecture is built upon deterministic recursive descent with zero dynamic FFI calls

```
// Geometric layout box dispatch in MoonBit
fn layout_element(ctx : StyleContext, box : LayoutBox) -> Unit {
  let x = box.rect.x
  let y = box.rect.y
  let w = box.rect.w
  let h = box.rect.h
  // Compute baseline alignment and font metrics
}
```

3. Multi-Target Rendering Pipelines

The compilation pipeline provides deterministic generation across three distinct vector targets

• SVG 1.1 Vector Graphics

Scalable XML graphics with font family definitions and path outlines

• Adobe PDF 1.4 Binary Documents

Self-contained indirect objects and binary font dictionaries

• Semantic HTML5

Clean document hierarchy with responsive styling and CSS grid compatibility

4. Benchmark and Evaluation

The benchmark in `benchmarks/pipeline_test.mbt` measures parse, layout, HTML, SVG, and PDF stages on a fixed input. Run it for each target under recorded host and toolchain conditions before